

Jiabao Hu

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EDUCATION

- **University of Southern California (USC)** · Los Angeles, CA Aug. 2023 – Dec. 2025
Master of Science in Computer Science GPA: 3.85/4
- **University of Illinois at Urbana-Champaign (UIUC)** · Urbana, IL Sep. 2019 – Jun. 2023
Bachelor of Science in Civil Engineering · with Honors · Minor in Computer Science GPA: 3.75/4.0
- **Zhejiang University (ZJU)** · Hangzhou, China Sep. 2019 – Jun. 2023
Bachelor of Engineering in Civil Engineering GPA: 3.93/4.0

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, Java, JavaScript, TypeScript, HTML, CSS, Shell
- **Technologies:** Spring Cloud, Spring Boot, Nginx, Angular, React, Flask, Django, Node.js, Express.js, Bootstrap, AWS, GCP, Azure, Heroku, Supabase, MongoDB, CI/CD, Android, OpenGL, WebGL, WebSocket, Figma, Tailwind
- **Tools:** MySQL, PostgreSQL, NPM, Redis, RabbitMQ, PyTorch, Git, Jupyter, Docker, Linux, WSL, VirtualBox, LaTeX
- **Certifications:** 🏆 Certified Scrum Developer

WORKING EXPERIENCE

- **Scrum Adventures - Full Stack Engineer Intern** Jun. 2025 - Present
Django, NextJS, React, Tailwind, PostgreSQL, Heroku, GitHub Actions, OpenAI API Long Beach, CA
 - Applied **Agile principles** in a **Scrum** team to modernize CaDoMi™ Tools, migrating from monolithic Django+HTML to a decoupled **NextJS frontend with REST API backend**, improving performance by **45%**.
 - Developed a scalable agile management platform serving **500+** users across 50+ teams, featuring sprint capacity planning, Kaizen tracking, and real-time metrics dashboards.
 - Built an **AI-powered Agile Coach** leveraging **OpenAI API**, delivering personalized retrospectives and guidance, achieving **92%** user satisfaction and **40%** improvement in velocity estimation.
 - Designed responsive **UI/UX** with stakeholder-driven research, boosting engagement by **55%** and reducing bounce rate by **30%**.
 - Enhanced **CI/CD** pipelines with **GitHub Actions**, ensuring zero-downtime releases across **15+** deployments and maintaining **99.5% uptime**.
- **Chuanzhi - Java Backend Developer Intern** Jun. 2025 - Aug. 2025
SpringCloud, MyBatis-Plus, MySQL, Redis, RabbitMQ, MongoDB, MinIO Remote
 - Built and maintained **SpringCloud microservices (Nacos + Gateway)** for a public-benefit rideshare delivery platform with **2,000+** active users.
 - Integrated **Baidu AI OCR** for license plate and vehicle registration verification, cutting manual review time by **80%**.
 - Implemented **RabbitMQ** delayed messaging to auto-cancel expired invitations, reducing stale orders by **70%**.
 - Developed a real-time chat system via **WebSocket** with **<200 ms** latency and **MongoDB** persistence to prevent data loss.
- **ZJU-UIUC Institute – Research Assistant** Sep. 2022 – Apr. 2023
Python, PyTorch, Blender, OpenCV, DeepLabV3+, ResNet-101 Haining, China
 - Engineered a modular **3D simulation platform** in Python/Blender, designing the "Random Bridge Generator" procedural modeling tool that programmatically builds synthetic bridge environments with **6 bridge types** and randomized structural geometry.
 - Built scalable **data generation pipelines** for CV model training, automating generation of **10K+ photorealistic images** with pixel-wise annotations using UAV-style camera views and Cityscapes overlays.
 - Integrated and trained **deep learning pipelines** using PyTorch, achieving **85.9% IoU** for column/pier detection and **79.8% IoU** for deck recognition with **DeepLabV3+ ResNet-101** architecture.
 - Published research findings in peer-reviewed journal, contributing to structural health monitoring and automated bridge inspection methodologies. [GitHub Link](#) [Publication Link](#)

SELECTED PROJECTS

- **Careenicle - Job Application Tracker**🏆: *Next.js, Vercel, Supabase, PostgreSQL, Tailwind CSS, Google Maps API, GenAI*
 - Built a modern job application tracking platform using **NextJS + Supabase** full-stack architecture, co-developed with **GenAI** for rapid prototyping, and deployed on **Vercel** with secure multi-user support and real-time synchronization.
 - Integrated **Google Maps API** for interactive job location visualization with custom markers and **Google Places autocomplete** for accurate geocoding.
 - Implemented comprehensive job management features including application tracking, interview scheduling, and event timelines with responsive design.
 - Leveraged **Supabase Auth with Row Level Security** to ensure complete data isolation and privacy between users.
- **Full-Stack Web Weather App**🏆: *Angular, Node.js, Express.js, MongoDB Atlas, AWS EC2, Tomorrow.io API*
 - Developed a responsive weather app with **Angular** and **Express.js**, supporting real-time forecasts with UI updates under **100ms**.
 - Designed scalable **RESTful APIs** to fetch live weather data from **Tomorrow.io**, sustaining **300ms average latency** under load.
 - Integrated **MongoDB Atlas** to persist up to **50+ favorite cities** per user across sessions with **zero data loss**.
 - Deployed on **AWS EC2** with Nginx reverse proxy, autoscaling to support **1000+ concurrent users** with 99.9% uptime.